1.List 5 real-world Indicators of Compromise (IoCs) that you might find on a system infected by malware, and for each, explain in one line how it could be detected in system or network logs.

No.	IoC	How to Detect It
1	Suspicious IP address	Check firewall logs for connections to known malicious IP addresses.
2	Unknown file or malware	Check antivirus logs for detected or quarantined files.
3	Unusual network traffic	Check network logs for large or unexpected data transfers.
4	Repeated failed logins	Check authentication logs for many failed login attempts from one IP address.
5	Suspicious domain name	Check DNS logs for connections to unusual or known malicious domains.

2.Install Splunk (or use Splunk Free Cloud Trial), upload a sample web server log file, and create a dashboard that shows the top 5 IP addresses accessing the server.

Steps:

1. Install Splunk or use an available Splunk trial.
2. Open Splunk and go to Add Data.
3. Upload a sample web server log file.
4. Select the correct source type, such as access_combined, if it matches your log format.
5. Open the Search & Reporting section.
6. Enter the following query:

```
index=main
| stats count by clientip
| sort - count
```

| head 5

7. If your log uses a different IP field, replace clientip with the correct field name.
8. Click Visualization and select a bar chart.
9. Save the chart to a dashboard with the title Top 5 IP Addresses.

**3.Using ELK Stack (Elasticsearch, Logstash, Kibana) or Splunk, write a query to detect failed login attempts from the same IP address within a 10-minute window.

Hint: Use aggregation or stats functions to group by IP and filter by event type.**

planation:

1. index=main searches the main log index.
2. "failed login" searches for failed login events.
3. bin _time span=10m groups events into 10-minute time windows.
4. stats count by src_ip, _time counts failed logins from each IP.
5. where count >= 5 shows IP addresses with five or more failed attempts in a window.

4.Simulate a suspicious activity by inserting a fake log entry that shows an unusual port scan (e.g., multiple connection attempts to different ports from one IP), and use Splunk or Kibana to identify and visualize this anomaly.

Step 1: Create a Sample Log Entry

Use the following fake log entries for your assignment:

2026-10-13 10:00:01 src_ip=192.168.1.50 dest_ip=192.168.1.10 dest_port=21 action=blocked

2026-10-13 10:00:02 src_ip=192.168.1.50 dest_ip=192.168.1.10 dest_port=22 action=blocked

2026-10-13 10:00:03 src_ip=192.168.1.50 dest_ip=192.168.1.10 dest_port=23 action=blocked

2026-10-13 10:00:04 src_ip=192.168.1.50 dest_ip=192.168.1.10 dest_port=80 action=blocked

2026-10-13 10:00:05 src_ip=192.168.1.50 dest_ip=192.168.1.10 dest_port=443 action=blocked

These are sample entries for testing, not real network events.

Step 2: Search in Splunk

After uploading the sample logs, run this query:

```
index=main
| stats dc(dest_port) as unique_ports
  count as total_attempts by src_ip
| where unique_ports >= 5
```

Step 3: Visualize the Results

1. Open the Visualization tab.
2. Select a bar chart.
3. Display the source IP and number of unique ports.
4. Save the chart as Suspicious Port Scan Detection.



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Expected Result: The IP address 192.168.1.50 should appear because it attempted connections to five different ports.